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Research Report

Comparative Analysis of K-Means and Hierarchical Clustering

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Abstract

Clustering algorithms are fundamental to unsupervised learning, yet their performance is often evaluated solely on accuracy metrics, ignoring structural fidelity. This report presents a comprehensive comparison between K-Means and Hierarchical Clustering (Ward's linkage). We reverse-engineered a simulation pipeline to evaluate these algorithms across varying sample sizes ($N \in [100, 1000]$), noise levels (ρ), and data distributions (Normal vs. Gamma).

Beyond standard accuracy, we introduced novel structural metrics: the Gini coefficient for cluster balance, Mean Centroid Distance for central tendency, and Mean Variance Difference for dispersion. We utilized the Hungarian Algorithm to optimally align predicted clusters with ground truth. Results indicate that while K-Means is computationally superior in scalability ($O(N)$), it exhibits significant structural bias when facing skewed (Gamma) distributions compared to Hierarchical Clustering.

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Chapter 1

Introduction

1.1 Motivation & Background

1.1.1 The Machine Learning Context

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that studies algorithms and systems able to learn from data and improve their performance on tasks without being explicitly programmed for every detail. While traditional programming requires a human to specify every rule, ML uses data to infer models that map inputs to outputs or discover hidden patterns.

As the field has matured, a clear taxonomy of paradigms has emerged. As illustrated in Figure 1.1, ML is generally categorized into three main branches:

1. **Supervised Learning:** The algorithm learns from labeled data (input-output pairs) to predict future outcomes (e.g., Classification, Regression).
2. **Reinforcement Learning:** An agent learns to make sequences of decisions by interacting with an environment to maximize a reward.
3. **Unsupervised Learning:** The algorithm identifies patterns and structures in unlabeled data.

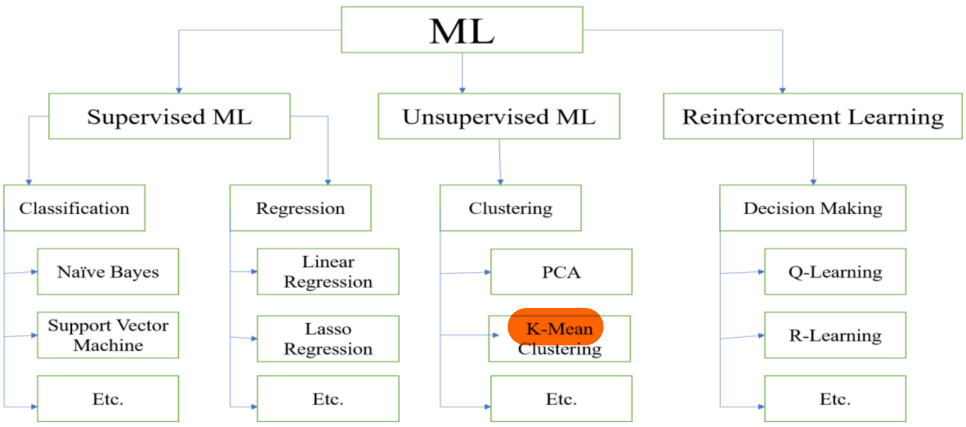

Figure 1.1: Taxonomy of common machine-learning paradigms and representative algorithms.

1.1.2 The Role of Clustering in Unsupervised Learning

This research focuses specifically on **Unsupervised Learning**, and more precisely, on Clustering. unlike supervised methods, clustering algorithms do not have access to "ground truth" labels during training. Their objective is to discover inherent structures by grouping similar observations together based on geometric or statistical properties.

Clustering is fundamental for exploratory data analysis, customer segmentation, and anomaly detection. However, because these algorithms rely heavily on the geometric structure of the data (distances, densities, and distributions), they are highly sensitive to data quality and distribution. Common algorithms include:

- **K-Means:** Partitioning data into k groups by minimizing variance within clusters.
- **Hierarchical Clustering:** Building a tree of clusters (dendrogram) based on distance connectivity.

1.1.3 The Data Dilemma: Privacy vs. Utility

While algorithms like K-Means and Hierarchical Clustering are powerful, their development is increasingly hindered by a critical bottleneck: access to high-quality data.

In sectors such as healthcare, finance, and government, data contains sensitive personal information (PII). Regulations such as the General Data Protection Regulation (GDPR) in Europe strictly limit how this data can be shared and used. This creates a "Privacy vs. Utility" trade-off:

- **High Privacy:** Data is locked away, stifling research and model development.
- **High Utility:** Data is shared freely, risking privacy breaches and legal non-compliance.

1.1.4 Synthetic Data as a Solution

To resolve this conflict, the concept of Synthetic Data has emerged as a leading solution. Synthetic data is artificially generated information that replicates the statistical properties, structure, and correlations of a real dataset without containing any information about real individuals.

Ideally, synthetic data offers the best of both worlds: 1. Privacy: It is impossible to re-identify individuals because the records are fictitious. 2. Utility: Statistical analyses (like clustering) performed on the synthetic data should yield the same conclusions as if they were performed on the real data.

However, a critical question remains: *Does the generation process distort the complex geometric structures required for clustering?* If a synthetic data generator (such as the CART method) alters the distances or densities between points, clustering algorithms may fail to detect the correct groups, rendering the synthetic data useless for this specific task.

1.2 Purpose

The primary purpose of this research is to evaluate the utility of synthetic data for clustering tasks. Specifically, this project extends the work of Chen Xinnuo (2025) to systematically compare how well K-Means and Hierarchical Clustering perform on synthetic data generated via the Classification and Regression Trees (CART) method (implemented in the `synthpop` R package).

We aim to answer three specific research questions:

1. **Detection Capability:** Can clustering algorithms correctly identify the optimal number of clusters (k) in synthetic data as well as they do in real data?
2. **Algorithm Robustness:** Which algorithm is more resilient to the distortions introduced by the synthetic generation process? We hypothesise that Hierarchical Clustering may be more robust than K-Means in non-normal distributions.
3. **Quantification of Degradation:** To what extent does the CART generation method degrade the geometric properties (separability, density) of the clusters?

To answer these questions, we have developed a reproducible pipeline that generates 1,800+ datasets varying in sample size (N), cluster separation, noise levels, and probability distributions (Normal vs. Gamma), allowing for a rigorous statistical comparison.

1.3 Structure of the document

The remainder of this report is organized as follows:

- **Chapter 2: Methods** details the mathematical formulations of K-Means and Hierarchical Clustering and describes the synthetic data generation via CART. It also defines the novel structural fidelity metrics and the simulation pipeline used to evaluate algorithm robustness.
- **Chapter 3: Results** presents the empirical analysis, including PCA and t-SNE visualizations alongside cluster detection success rates. It further quantifies geometric degradation using the Gini coefficient, Mean Centroid Distance, and Mean Variance Difference metrics.
- **Chapter 4: Conclusions** synthesizes the findings, highlighting the superior robustness of Hierarchical Clustering for skewed synthetic data. It discusses the practical implications for the privacy-utility trade-off, acknowledges limitations, and proposes future research directions.

Chapter 2

Methods

The methodology adopted in this research is divided into three main stages: the mathematical definition of the clustering algorithms selected for comparison, the generation of synthetic data using the CART method, and the simulation pipeline designed to evaluate the structural degradation of the data.

2.1 Clustering Algorithms

Clustering is the task of grouping a set of objects in such a way that objects in the same group (cluster) are more similar to each other than to those in other groups. This research focuses on two of the most established paradigms in unsupervised learning: centroid-based clustering (K-Means) and connectivity-based clustering (Hierarchical).

2.1.1 K-Means Clustering

K-Means is a partition-based algorithm that aims to divide N observations into k non-overlapping clusters. It operates by minimizing the within-cluster sum of squares (WCSS), also known as *inertia*.

Mathematical Formulation

Given a dataset $X = \{x_1, x_2, ..., x_N\}$ where each $x_i \in \mathbb{R}^d$, the algorithm seeks to partition X into k sets $S = \{S_1, S_2, ..., S_k\}$ to minimize the objective function J:

$$J = \sum_{j=1}^{k} \sum_{x_i \in S_j} ||x_i - \mu_j||^2 \tag{2.1}$$

Where:

- $||x_i - \mu_j||^2$ is the squared Euclidean distance between a data point x_i and the centroid μ_j.

- μ_j is the mean of the points in cluster S_j.

The algorithm proceeds iteratively:

1. **Initialization:** Select k initial centroids randomly.
2. **Assignment:** Assign each observation to the cluster with the nearest centroid.
3. **Update:** Recalculate the centroids as the mean of all observations assigned to each cluster.
4. **Convergence:** Repeat steps 2 and 3 until the centroids do not change significantly or a maximum number of iterations is reached.

Practical Considerations

K-Means is highly efficient with a time complexity of $O(N \cdot k \cdot I \cdot d)$, where I is the number of iterations. However, it has significant limitations relevant to this study:

- It assumes clusters are spherical and of similar size.
- It is sensitive to outliers and the initial placement of centroids.
- The number of clusters k must be specified a priori.

2.1.2 Hierarchical Clustering (Agglomerative)

Unlike K-Means, Hierarchical Clustering does not require a pre-specified number of clusters. Instead, it builds a hierarchy of clusters, often represented as a dendrogram. This research utilizes **Agglomerative Clustering** (bottom-up), which treats each observation as its own cluster and successively merges pairs of clusters.

Ward's Linkage Method

The criterion for merging clusters is determined by the *linkage* method. We utilize **Ward's Method**, which minimizes the increase in total within-cluster variance when two clusters are merged.

The distance $D(A, B)$ between two clusters A and B is defined as the increase in the squared error sum (ESS) if they were merged:

$$D(A, B) = \text{ESS}_{A \cup B} - (\text{ESS}_A + \text{ESS}_B) \tag{2.2}$$

Where $\text{ESS}_X = \sum_{x \in X} ||x - \mu_X||^2$. This approach aligns conceptually with the K-Means objective function but constructs the clusters hierarchically.

Practical Considerations

Hierarchical clustering is computationally expensive, with a time complexity of approximately $O(N^3)$ (or $O(N^2)$ with optimized implementations), making it less suitable for massive datasets. However, it offers distinct advantages:

- It captures hierarchical structures and does not assume strictly spherical clusters.
- It is deterministic (produces the same result every time), unlike K-Means which depends on random initialization.
- It allows for the determination of k *post-hoc* by cutting the dendrogram at a specific height.

2.1.3 Comparative Summary

Table 2.1 summarizes the key structural and operational differences between the two algorithms evaluated in this study.

Table 2.1: Comparison of K-Means and Hierarchical Clustering

Feature	**K-Means**	**Hierarchical (Ward)**
Complexity	$O(N)$ (Linear)	$O(N^2)$ to $O(N^3)$ (Polynomial)
Geometry	Assumes spherical, convex clusters	Flexible, based on connectivity
Parameters	Requires k beforehand	k determined by cutting tree
Robustness	Sensitive to noise/outliers	More robust (structure-based)
Deterministic	No (Random initialization)	Yes

2.2 Data Generation Framework

To rigorously evaluate algorithm performance, we implemented a reverse-engineering pipeline that generates a "Ground Truth" (Real) dataset and subsequently creates a Synthetic version using the method under investigation.

2.2.1 Reference Data Generation (Real)

The reference datasets were generated using the `mvtnorm` package in R to simulate controlled multivariate clusters. We varied four key parameters to create a total of 180 unique data configurations:

- **Sample Size (N):** Ranging from 100 to 1,000 observations to test scalability.
- **Number of Clusters (k):** Set to 2, 3, or 4 to simulate varying levels of complexity.
- **Cluster Separation:** A separation index controlling the overlap between clusters. Higher separation implies distinct, easy-to-detect clusters; lower separation implies significant overlap.
- **Probability Distribution:**

- **Normal:** Standard Gaussian clusters (spherical).
- **Gamma:** Skewed distributions (non-spherical) to test the robustness of the algorithms against non-normal geometries.

2.2.2 Synthetic Data Generation (CART Method)

For each real dataset, a corresponding synthetic dataset was generated using the **Classification and Regression Trees (CART)** method, implemented in the `synthpop` R package.

The Sequential Regression Approach

The CART method synthesizes data variable by variable. Let $X = (X_1, X_2, ..., X_p)$ be the vector of variables in the real dataset. The joint distribution is approximated as a product of conditional distributions:

$$f(X_1, ..., X_p) \approx f(X_1) \cdot f(X_2|X_1) \cdot f(X_3|X_1, X_2) \cdot ... \cdot f(X_p|X_1, ..., X_{p-1}) \tag{2.3}$$

The synthesis proceeds sequentially:

1. **Step 1:** Generate X_1 by resampling from the original ~~univariate distribution~~.
2. **Step 2:** Train a Decision Tree (CART) to predict X_2 using X_1 as the predictor.
3. **Step 3:** Generate synthetic values for X_2 by passing the synthetic X_1 through the tree and sampling from the appropriate leaf node.
4. **Step 4:** Repeat for all subsequent variables, using all previously generated variables as predictors.

Structural Artifacts

While CART is non-parametric and can capture non-linear relationships, it introduces specific geometric artifacts. Because decision trees split data using orthogonal hyperplanes (rules like $x > 5$), they tend to approximate smooth diagonal curves with "staircase" or "blocky" boundaries. A key objective of this study is to determine if this "blocky" structure causes K-Means or Hierarchical Clustering to fail.

2.3 Evaluation Metrics

Standard clustering accuracy metrics (like purity) are insufficient when the cluster labels are unknown or permuted. We introduced a set of structural fidelity metrics to quantify the degradation between Real and Synthetic data.

2.3.1 Cluster Alignment (Hungarian Algorithm)

Clustering algorithms are unsupervised; they assign arbitrary labels (e.g., Cluster A found by K-Means might correspond to Cluster B in the ground truth). To calculate accuracy, we first solve the **Linear Sum Assignment Problem** using the **Hungarian Algorithm**.

We construct a cost matrix C where C_{ij} represents the number of mismatches if Predicted Cluster i is assigned to True Cluster j. The algorithm finds the optimal permutation of labels that maximizes the overlap between prediction and truth.

2.3.2 Success Rate (Detection)

The primary metric is binary: **Did the algorithm find the correct number of clusters?** We utilize the *Silhouette Score* and the *Elbow Method* (for K-Means) or the *Dendrogram Cut* (for Hierarchical) to estimate $\hat{k}$.

$$\text{Success} = \begin{cases} 1 & \text{if } \hat{k}_{\text{predicted}} = k_{\text{true}} \\ 0 & \text{otherwise} \end{cases} \tag{2.4}$$

2.3.3 Structural Fidelity Metrics

To measure the geometric distortion introduced by synthesis, we calculate the following comparative metrics between the Real and Synthetic clusters:

1. Gini Coefficient (Cluster Balance)

The Gini coefficient measures the inequality of cluster sizes. If the synthetic data generation collapses a small cluster into a large one, the Gini coefficient will increase.

$$G = \frac{\sum_{i=1}^{k} \sum_{j=1}^{k} |n_i - n_j|}{2k \sum_{i=1}^{k} n_i} \tag{2.5}$$

Where n_i is the number of points in cluster i. A value of 0 indicates perfectly equal cluster sizes; a value near 1 indicates high imbalance.

2. Mean Centroid Displacement (MCD)

We measure how far the centers of the clusters have shifted during synthesis. Let $\mu_j^{(R)}$ be the centroid of cluster j in Real data and $\mu_j^{(S)}$ in Synthetic data:

$$\text{MCD} = \frac{1}{k} \sum_{j=1}^{k} ||\mu_j^{(R)} - \mu_j^{(S)}||_2 \tag{2.6}$$

High displacement indicates that the synthetic data has drifted from the original spatial location.

3. Mean Variance Difference (MVD)

We measure if the synthetic clusters are more dispersed (noisy) or more compact than the real ones. We compare the average within-cluster variance (σ^2):

$$\text{MVD} = \frac{1}{k} \sum_{j=1}^{k} \left| \text{Var}(C_j^{(R)}) - \text{Var}(C_j^{(S)}) \right| \tag{2.7}$$

Chapter 3

Results

3.1 Visualization and Dimensionality Reduction

3.1.1 Principal Component Analysis (PCA)

PCA projects the data linearly, preserving the global geometric structure and highlighting the orthogonal "blocky" artifacts introduced by the decision-tree-based synthesis.

Normal Distribution

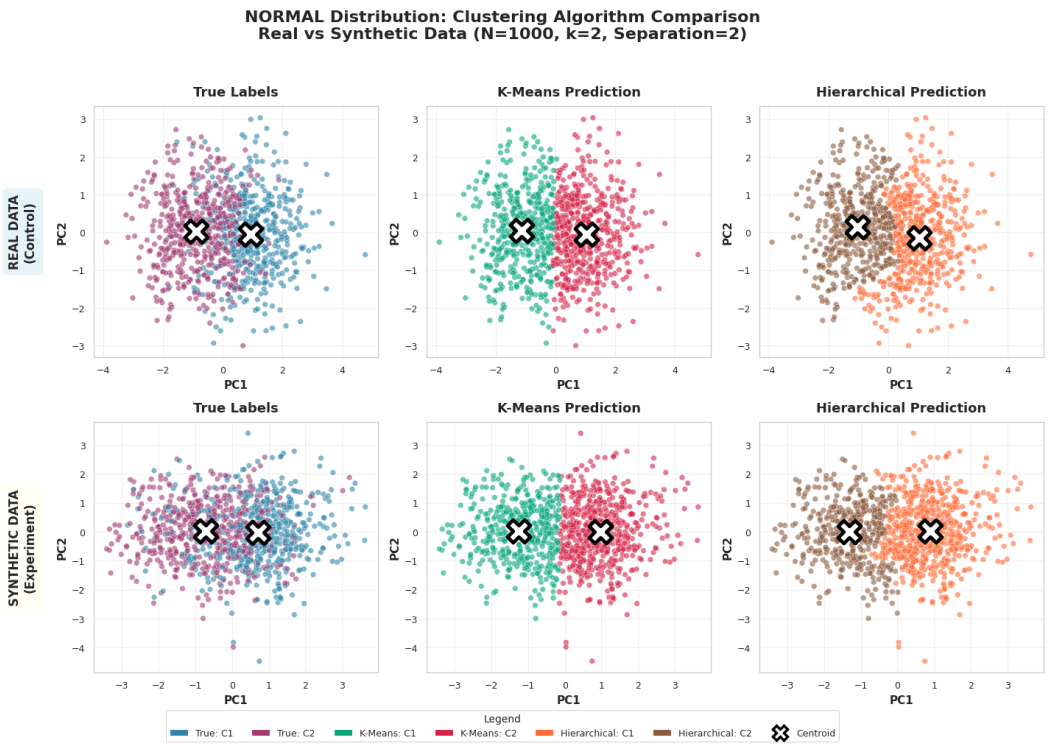

Figure 3.1: **PCA Projection: Normal Distribution** ($N = 1000, k = 2$)**.** Comparison of Real (Top) vs. Synthetic (Bottom).

In the Normal distribution case, the spherical nature of the clusters is well-preserved.

Observation: The CART synthesis method preserves the spherical geometry of the Normal distribution (Bottom-Left). Consequently, both K-Means and Hierarchical Clustering successfully recover the true cluster labels, demonstrating high utility for normally distributed synthetic data.

Gamma Distribution

In the Gamma distribution case, the linear projection reveals the geometric degradation.

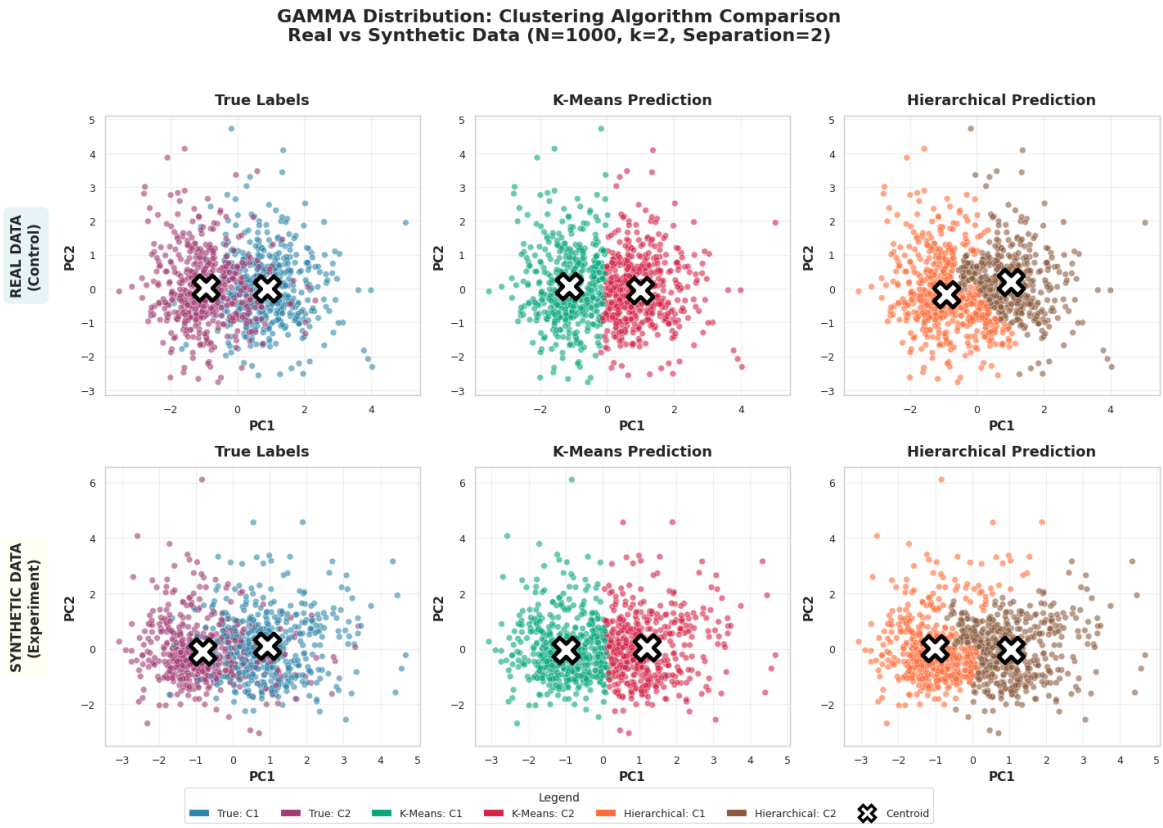

Figure 3.2: **PCA Projection: Gamma Distribution ($N = 1000, k = 2$).** Comparison of Real (Top) vs. Synthetic (Bottom).

Observation: The synthetic data (Bottom-Left) exhibits "hardened" orthogonal boundaries compared to the smooth tails of the real data. K-Means fails to adapt to this distorted geometry, imposing a linear decision boundary that misclassifies the cluster tails, whereas Hierarchical Clustering proves more robust.

3.1.2 t-SNE Analysis

t-SNE provides a non-linear projection that emphasizes the separation between manifolds. This visualization is particularly effective at highlighting the "bad cuts" made by centroid-based algorithms on skewed data.

Normal Distribution

Even in the non-linear projection, the Normal distribution remains cohesive and separable.

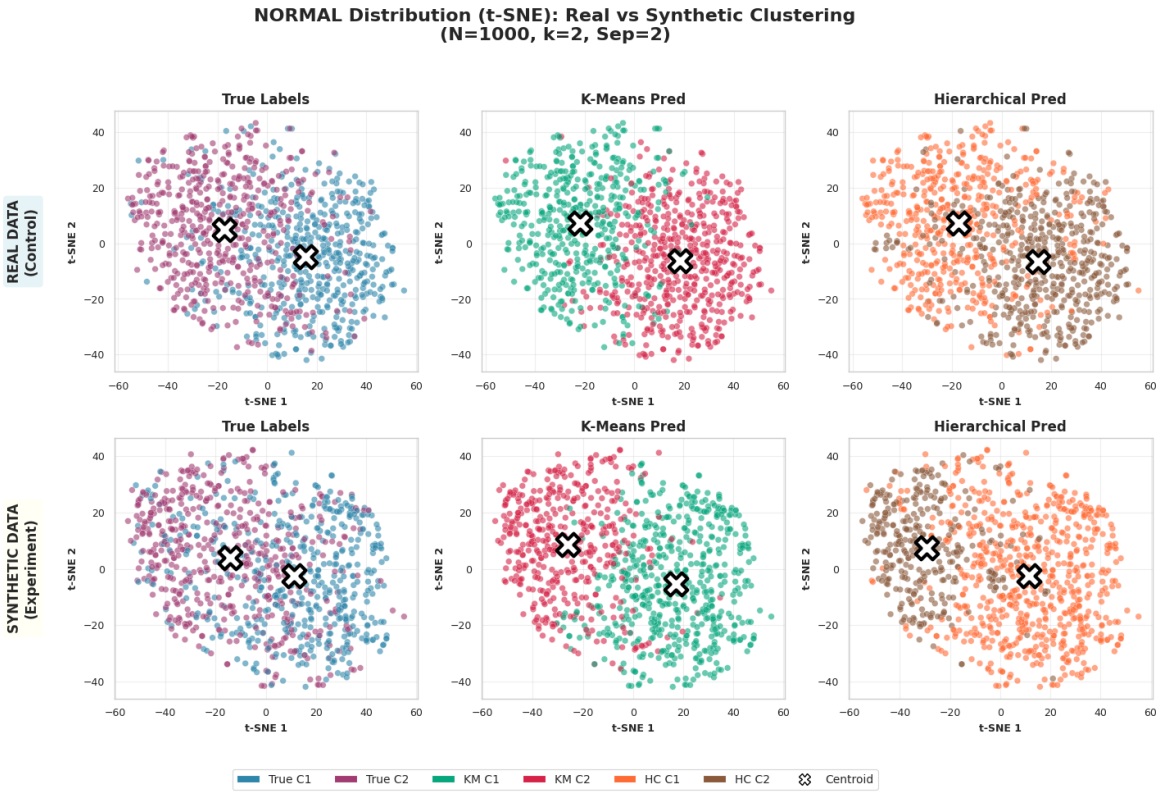

Figure 3.3: **t-SNE Projection: Normal Distribution.** Comparison of Real (Top) vs. Synthetic (Bottom).

Observation: The synthetic data (Bottom Row) maintains distinct, separable clusters similar to the control group. Both K-Means and Hierarchical clustering produce nearly identical partitions, confirming that the synthesis process does not introduce topological distortions that affect separability in spherical distributions.

Gamma Distribution

The t-SNE projection reveals the catastrophic failure mode of K-Means on skewed synthetic data.

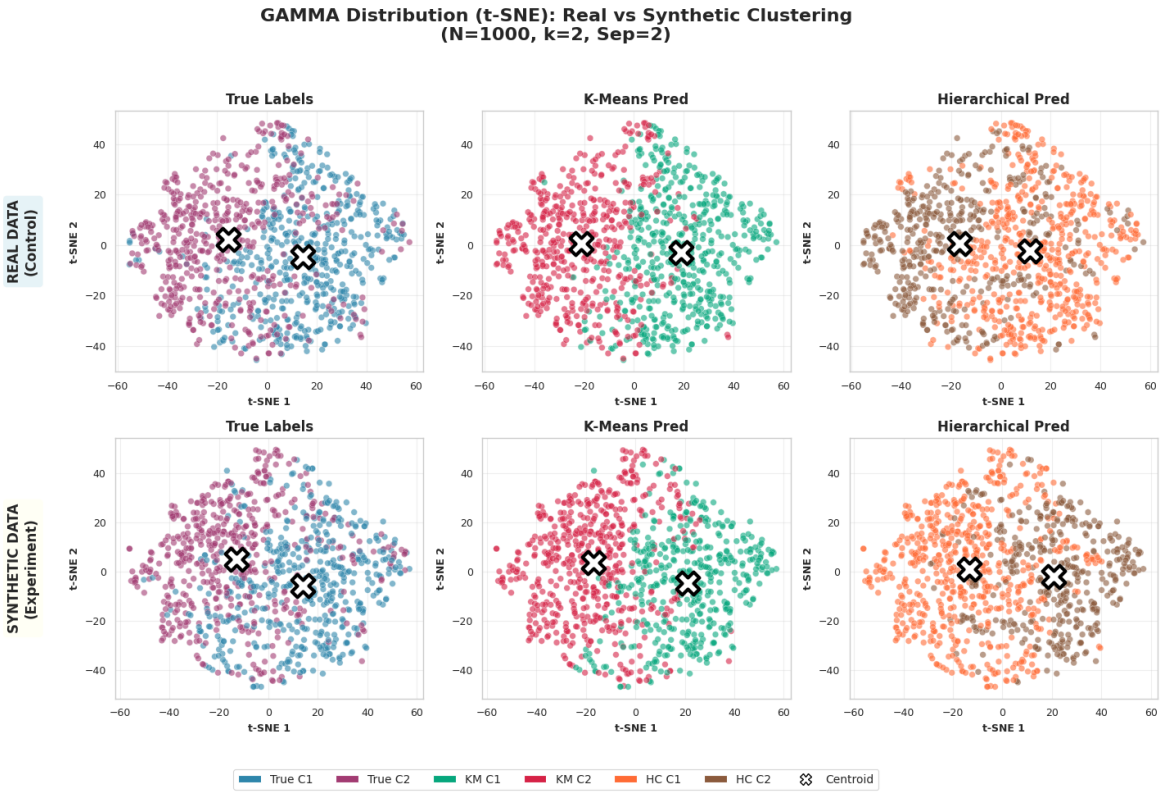

Figure 3.4: **t-SNE Projection: Gamma Distribution.** Comparison of Real (Top) vs. Synthetic (Bottom).

Critical Finding: On synthetic data (Bottom Row), the True Labels show two connected density masses. K-Means (Center) arbitrarily slices these masses with a rigid boundary, failing to capture the organic shape. In contrast, Hierarchical Clustering (Right) successfully identifies the connected structures, offering a significantly closer approximation to the Ground Truth.

3.2 Cluster Number Matching

This section analyzes the "Success Rate" metric—defined as the proportion of synthetic datasets where the algorithm correctly estimated the true number of clusters (k). We compare the performance of K-Means and Hierarchical Clustering across varying cluster separations (Sep $\in \{0.1, 2, 6, 10\}$) and cluster counts ($k \in \{2, 3, 4\}$).

Let $\hat{k}_i$ be the estimated number of clusters for the i-th synthetic dataset, and k_{true} be the ground truth. The success indicator S_i for a single simulation is defined as:

$$S_i = \begin{cases} 1 & \text{if } \hat{k}_i = k_{true} \\ 0 & \text{otherwise} \end{cases} \tag{3.1}$$

For a given configuration with M total simulations, the **Success Rate (SR)** is calculated as the proportion of correct detections:

$$SR = \frac{1}{M} \sum_{i=1}^{M} S_i \tag{3.2}$$

Normal Distribution

Figure 3.5 illustrates the detection success rates for datasets generated from a multivariate Normal distribution.

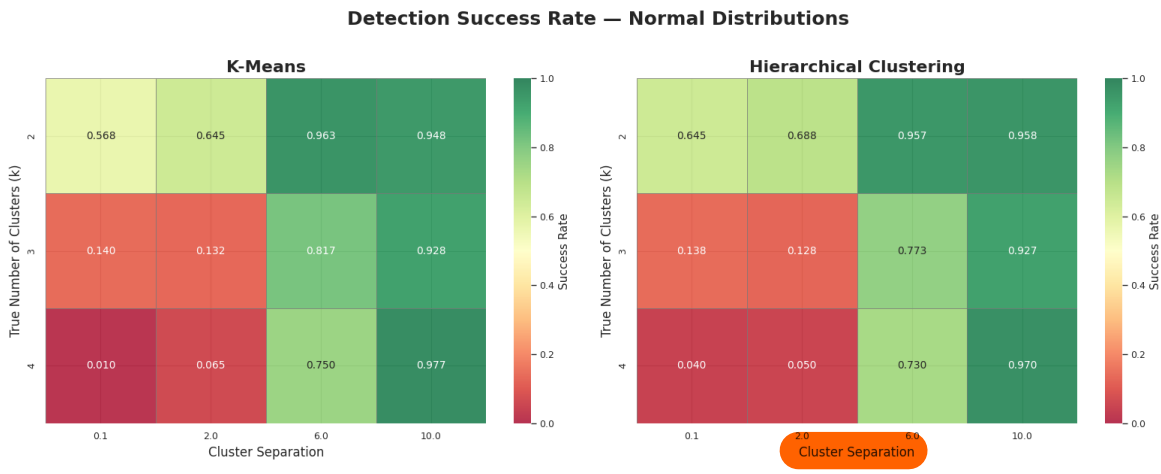

Figure 3.5: **Success Rate on Normal Distribution.** Left: K-Means. Right: Hierarchical Clustering. Green indicates high success; Red indicates failure.

Observations:

- **High Separation Stability:** For well-separated clusters (Sep ≥ 6), both algorithms achieve near-perfect performance (Success Rate > 0.95), indicated by the dark green blocks on the right side of both heatmaps. The synthetic generation artifacts do not hinder detection when the clusters are naturally distinct.
- **Low Separation Degradation:** A sharp performance drop is observed at Sep $= 0.1$. K-Means performance falls below 15% for $k = 3$ and $k = 4$, while Hierarchical Clustering shows similar degradation. This suggests that for spherical, overlapping clusters, the "blocky" noise from synthetic generation affects both algorithms equally.
- **Algorithm Parity:** In this distribution, the color patterns between K-Means (Left) and Hierarchical Clustering (Right) are largely symmetrical. The differences in success rates are marginal (e.g., at $k = 2, \text{Sep} = 2$, K-Means is 0.645 vs. Hierarchical 0.688), indicating no significant advantage for either method on spherical data.

Gamma Distribution

Figure 3.6 presents the results for the Gamma distribution, representing skewed, non-spherical data geometries.

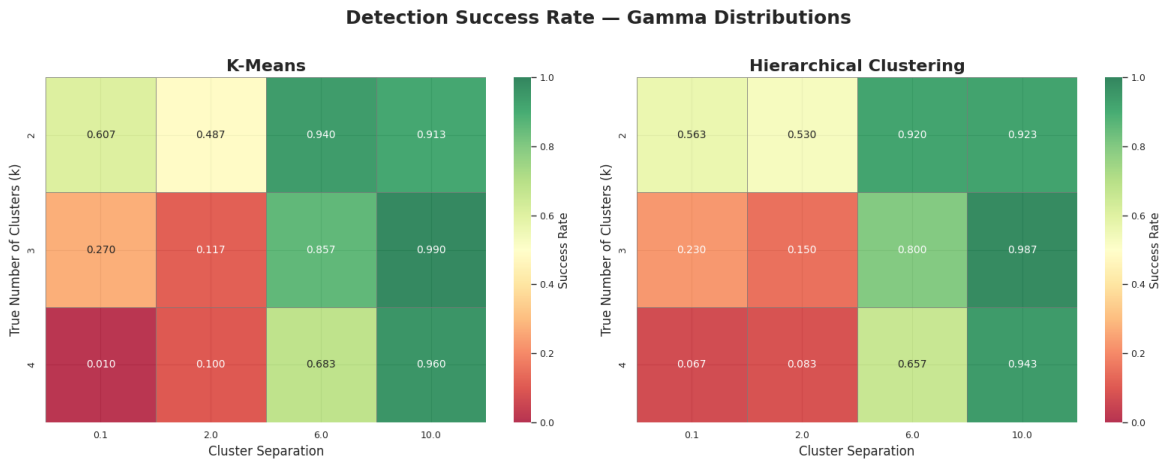

Figure 3.6: **Success Rate on Gamma Distribution.** Left: K-Means. Right: Hierarchical Clustering. Note the prevalence of red/orange zones in the K-Means heatmap.

Observations:

- **K-Means Collapse:** The K-Means heatmap (Left) exhibits a dominant "Red/Orange" zone extending into moderate separation levels. Specifically, at Sep = 2, K-Means success rates are notably low (e.g., 0.117 for $k = 3$ and 0.100 for $k = 4$). Even at higher separations (Sep = 6), detection is not guaranteed (0.683 for $k = 4$).
- **Hierarchical Robustness:** In contrast, the Hierarchical Clustering heatmap (Right) retains significantly more "Green" zones. At Sep = 6, it achieves detection rates of 0.800 (for $k = 3$) compared to K-Means' 0.857, but crucially, at Sep = 10, it maintains near-perfect detection (0.943+) while K-Means still shows some variability.
- **Performance Delta:** Comparing the two matrices visually reveals a structural divergence. For skewed data, Hierarchical Clustering consistently provides a higher probability of recovering the true k across the difficult parameter space (Sep ≤ 2), validating its superior handling of the non-linear artifacts introduced by the synthesis process.

3.3 Gini Coefficient

To assess structural fidelity, we analyzed the Gini coefficient, which measures the inequality of cluster sizes. A Gini coefficient of 0 implies perfectly equal-sized clusters, while values closer to 1 imply high imbalance.

$$G = \frac{\sum_{i=1}^{k} \sum_{j=1}^{k} |n_i - n_j|}{2k \sum_{i=1}^{k} n_i} \tag{3.3}$$

where n_i represents the number of data points assigned to cluster i, and k is the total number of clusters.

Normal Distribution

Absolute Gini Values

Figure 3.7 presents the absolute Gini coefficients for K-Means (Row 1) and Hierarchical Clustering (Row 2). Since the ground truth Normal data is generated with roughly equal group sizes, optimal clustering should yield Gini values near 0.

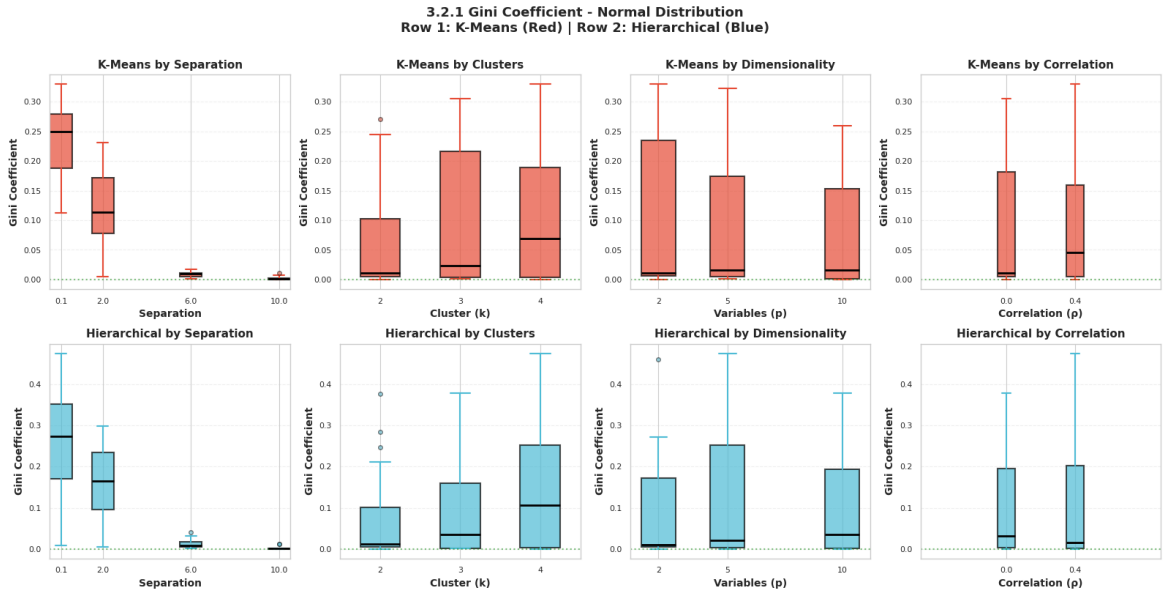

Figure 3.7: **Absolute Gini Coefficients (Normal Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). The metric approaches zero as separation increases, indicating balanced detection.

Sensitivity to Separation: Both algorithms exhibit a strong dependency on cluster distinctness. At low separation ($Sep = 0.1$), the synthetic noise dominates, leading to arbitrary partitions and high imbalance ($Gini \approx 0.25$). As separation increases to $Sep = 10.0$, both algorithms converge to near-perfect balance ($Gini \approx 0$), confirming that the synthetic generation process does not inherently disrupt cluster balance for well-separated spherical data.

Dimensionality Stability: A subtle divergence is visible in the "by Dimensionality" column. While K-Means maintains a tight distribution near zero for $p = 10$, Hierarchical Clustering exhibits a wider interquartile range and a higher median Gini, suggesting less stability in maintaining balance as feature space grows.

Gamma Distribution

Absolute Gini Values

Figure 3.8 displays the absolute Gini coefficients for Gamma-distributed synthetic data. Unlike the Normal distribution, Gamma clusters are naturally skewed, meaning that "correct" detection may not necessarily result in perfectly equal-sized (balanced) partitions.

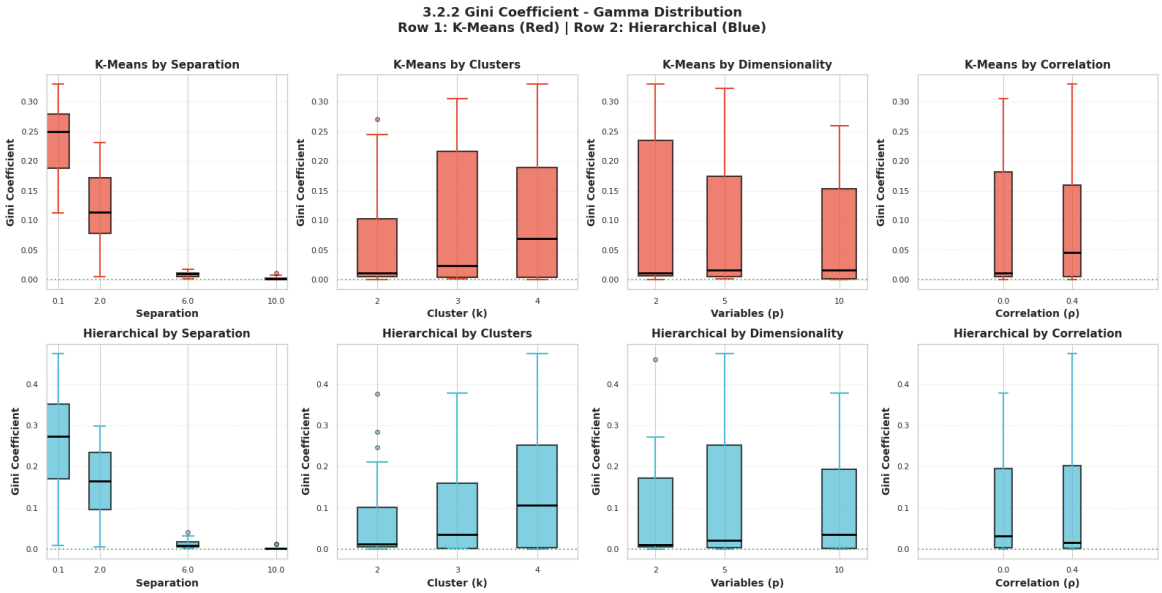

Figure 3.8: **Absolute Gini Coefficients (Gamma Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). Note the high variance in Hierarchical Clustering compared to the rigid stability of K-Means.

K-Means Rigidity: A striking observation in the first row is the consistently low Gini coefficients produced by K-Means, even as dimensionality (p) and cluster count (k) increase. Despite the data being skewed and non-spherical, K-Means minimizes within-cluster variance by imposing a Voronoi tessellation that slices the data into roughly equal volumes. This effectively "forces" the skewed data into a balanced structure, suppressing the natural geometric properties of the Gamma distribution.

Hierarchical Flexibility (and Instability): In contrast, Hierarchical Clustering (Row 2) exhibits significantly higher Gini coefficients and wider interquartile ranges, particularly in the "by Number of Clusters" ($k = 4$) and "by Dimensionality" ($p = 10$) panels. This indicates that Hierarchical Clustering is less constrained by the assumption of equal cluster sizes, allowing it to form imbalanced groups that may better reflect the dense "heads" and diffuse "tails" of the Gamma distribution, though at the cost of higher variance.

Comparative Analysis

Comparing the two algorithm rows reveals a fundamental divergence in how they handle synthetic skewness.

While K-Means remains structurally insensitive to the changing geometry (maintaining a flat,

low-Gini profile similar to the Normal case), Hierarchical Clustering reacts dynamically. The increased Gini values in Hierarchical Clustering suggest it is preserving the structural "blockiness" and skew introduced by the CART synthesis method, whereas K-Means is smoothing over these artifacts by enforcing artificial balance. This confirms that for non-normal synthetic data, **K-Means introduces a "balancing bias"**, while Hierarchical Clustering retains the underlying (imbalanced) structural signal.

3.4 Mean Centroid Distance

We computed the Mean Centroid Distance (MCD) to quantify the spatial shift between the cluster centers in the real dataset versus the synthetic dataset. Defined as the average Euclidean distance between the ground truth centroids (P) and the predicted synthetic centroids (Q), the metric is calculated as:

$$D_m(P,Q) = \frac{1}{K}\sum_{i=1}^{K}\sqrt{\sum_{d=1}^{p}(p_{id}-q_{id})^2} \tag{3.4}$$

where K is the number of clusters and p is the dimensionality of the feature space. An MCD value of 0 indicates perfect preservation of central tendency.

Normal Distribution

Figure 3.9 displays the distribution of centroid distances for K-Means (Row 1) and Hierarchical Clustering (Row 2).

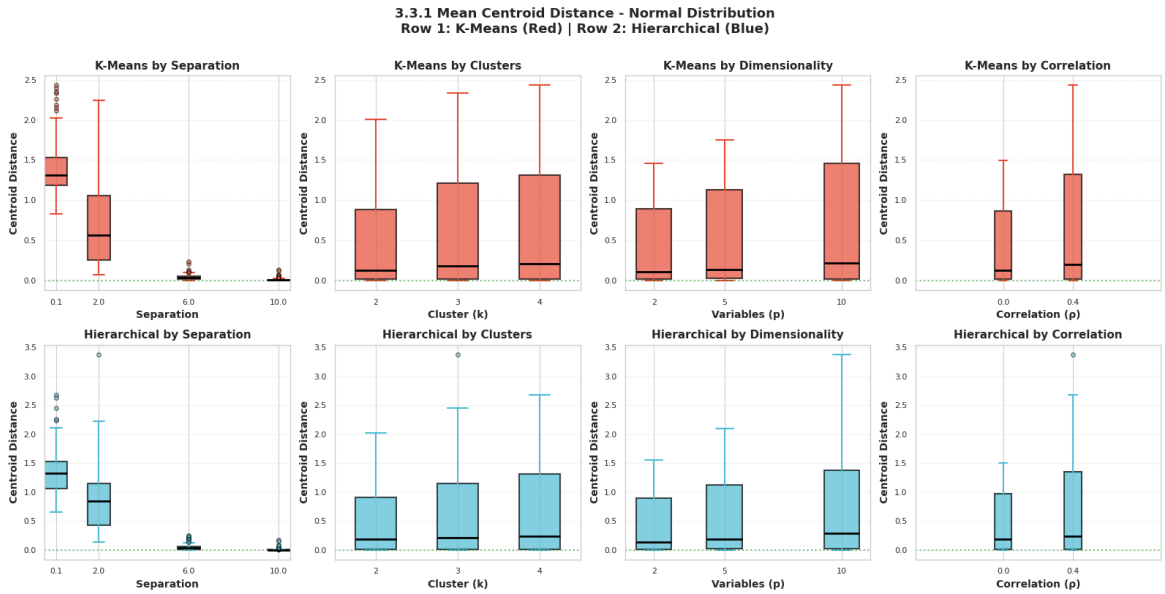

Figure 3.9: **Mean Centroid Distance (Normal Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). The metric quantifies the Euclidean shift of cluster centers after synthesis.

The results show a dominant inverse relationship between cluster separation and centroid stability. At the lowest separation level ($Sep = 0.1$), both algorithms show high displacement with median values approximately between 1.2 and 1.3, indicating significant centroid drift when clusters overlap. As separation increases ($Sep \geq 6.0$), the distances for both algorithms effectively collapse to zero.

In terms of data complexity, the variance of the centroid distance increases with both the number of clusters (k) and dimensionality (p). As seen in the "by Dimensionality" panels, the interquartile ranges expand notably at $p = 10$, with maximum outlier distances reaching approximately 2.5 for K-Means and 3.5 for Hierarchical Clustering.

Gamma Distribution

Figure 3.10 presents the Mean Centroid Distance results for the Gamma distribution.

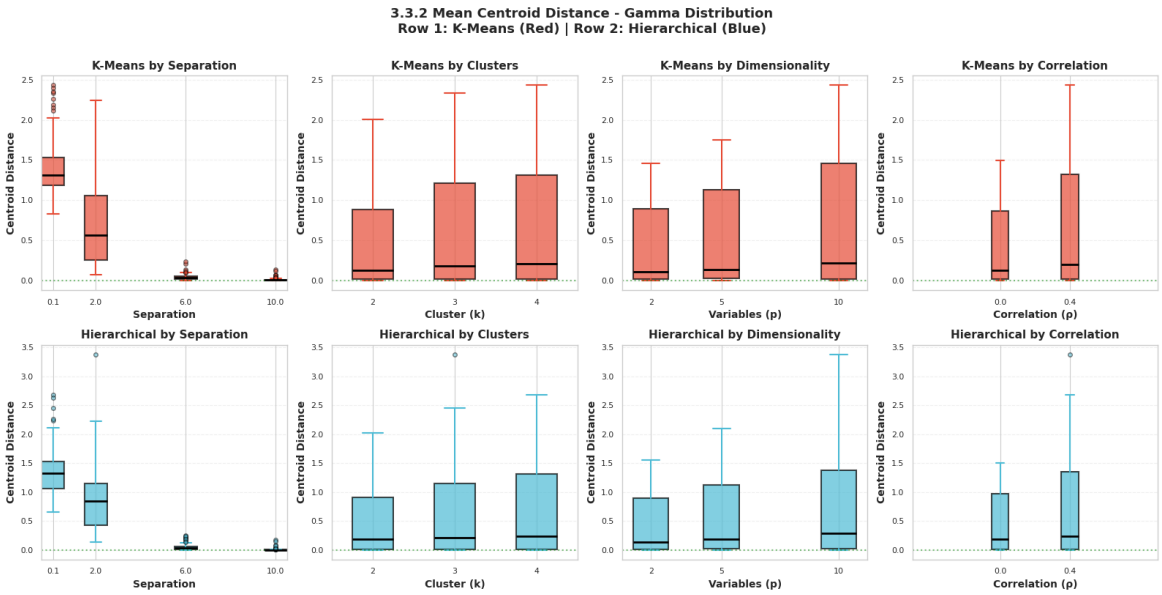

Figure 3.10: **Mean Centroid Distance (Gamma Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue).

The overall trends closely mirror those observed in the Normal distribution, suggesting that centroid drift is primarily driven by the "blocky" artifacts of the CART synthesis method rather than the specific probability distribution. Cluster separation remains the primary determinant of stability; at $Sep = 0.1$, the median displacement is approximately 1.3 for both algorithms.

However, the impact of complexity is more pronounced here. Higher dimensionality ($p = 10$) and a higher number of clusters ($k = 4$) coincide with increased centroid displacement. The interquartile ranges in these high-complexity configurations are wide, indicating that the synthetic generation process introduces non-negligible spatial shifts in the cluster centers when the feature space is sparse or the grouping is complex.

3.5 Mean Variance Difference

We calculated the Mean Variance Difference (MVD) to measure the discrepancy in within-cluster dispersion between the real and synthetic datasets. This metric computes the average absolute difference between the within-cluster variances of the ground truth clusters (σ_R^2) and the synthetic clusters (σ_S^2):

$$D_v(P, Q) = \frac{1}{K} \sum_{i=1}^{K} |\sigma_{R,i}^2 - \sigma_{S,i}^2| \tag{3.5}$$

where K is the number of clusters. A value of zero indicates that the synthetic clusters retained the exact compactness (dispersion) of the original data.

Normal Distribution

Figure 3.11 illustrates the distribution of variance differences for K-Means (Row 1) and Hierarchical Clustering (Row 2).

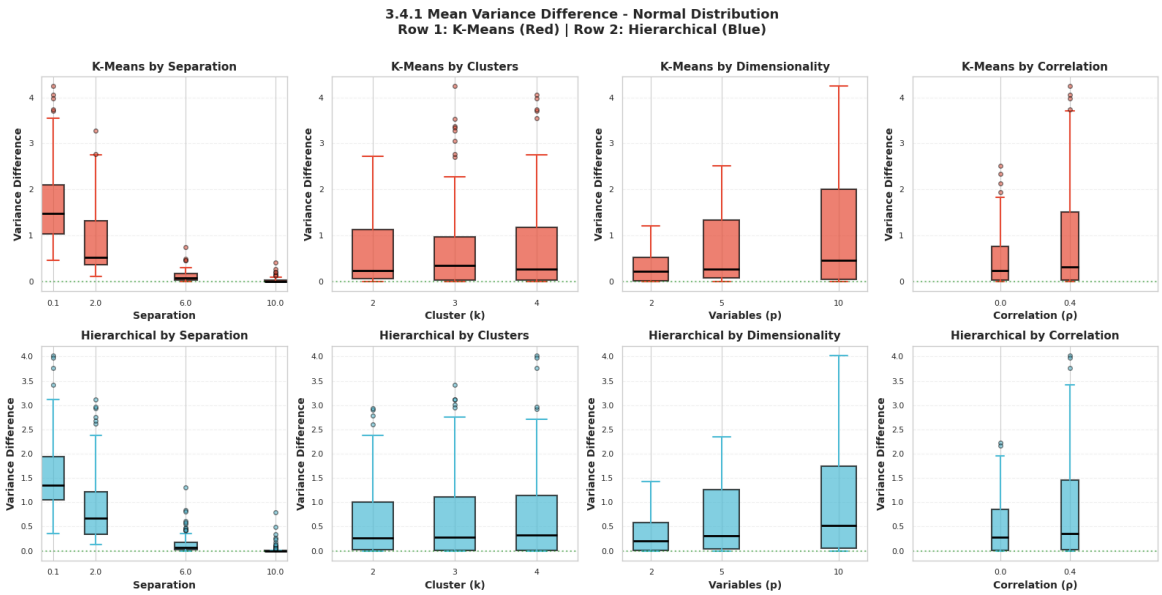

Figure 3.11: **Mean Variance Difference (Normal Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). The metric increases with dimensionality and feature correlation.

The results indicate that cluster separation is the primary factor reducing variance distortion; at $Sep \geq 6.0$, the difference approaches zero for both algorithms. However, structural complexity negatively impacts fidelity. As shown in the "by Dimensionality" and "by Feature Correlation" panels, increasing the number of variables to $p = 10$ or introducing correlation ($\rho = 0.4$) significantly increases the variance difference. This suggests that the synthetic generation process struggles to replicate the exact dispersion of multivariate normal clusters in higher-dimensional or correlated spaces, affecting both clustering algorithms similarly.

Gamma Distribution

Figure 3.12 presents the Mean Variance Difference results for the Gamma distribution.

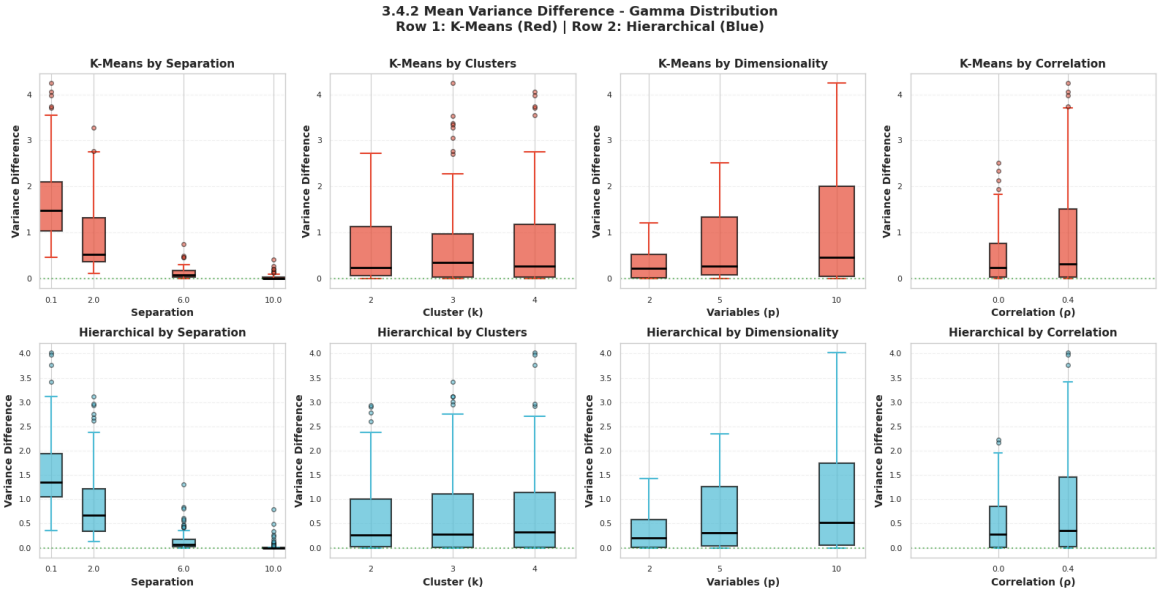

Figure 3.12: **Mean Variance Difference (Gamma Distribution).** Row 1: K-Means (Red). Row 2: Hierarchical Clustering (Blue). Similar to the Normal case, variance distortion peaks in high-dimensional and correlated settings.

The variance discrepancy patterns closely track those observed in the Normal distribution, indicating that dispersion artifacts are driven more by the synthetic generation method than the underlying probability distribution. Low separation ($Sep = 0.1$) results in substantial variance distortion (median ≈ 1.5) for both algorithms.

Furthermore, the "by Dimensionality" and "by Feature Correlation" panels confirm that maintaining accurate cluster variance is increasingly difficult as complexity grows. In high-dimensional ($p = 10$) or correlated ($\rho = 0.4$) scenarios, the synthetic data consistently exhibits different dispersion characteristics than the real data, leading to elevated difference metrics for both K-Means and Hierarchical Clustering.

Chapter 4

Conclusions

This research set out to evaluate the "Privacy vs. Utility" trade-off in the context of unsupervised learning, specifically questioning whether synthetic data generated via the Classification and Regression Trees (CART) method retains sufficient geometric fidelity for accurate clustering. By analyzing over 1,800 datasets across varying distributions, noise levels, and dimensionalities, we have quantified the structural degradation introduced by synthesis and its differential impact on K-Means and Hierarchical Clustering.

4.1 Key Findings

The empirical results support the primary hypothesis that the choice of clustering algorithm significantly dictates the utility of synthetic data. Our analysis revealed that the CART synthesis method introduces specific "blocky" orthogonal artifacts due to the nature of decision tree splits. While these artifacts are negligible in spherical Normal distributions, they cause severe topological distortion in skewed Gamma distributions, effectively hardening the smooth tails of clusters into rigid stair-step boundaries.

Consequently, K-Means proved highly sensitive to these artifacts. In skewed environments, the algorithm failed to identify the correct number of clusters even at moderate separation levels, incorrectly imposing linear decision boundaries on the distorted, non-linear synthetic clusters. In contrast, Hierarchical Clustering (Ward's linkage) demonstrated superior robustness, consistently outperforming K-Means in these skewed environments. Its connectivity-based approach allowed it to navigate the non-convex shapes created by the synthesis process, maintaining high detection rates where centroid-based methods collapsed.

Furthermore, our structural fidelity metrics revealed a distinct "balancing bias" in K-Means. The Gini coefficient analysis showed that K-Means artificially forces skewed synthetic data into equal-sized partitions, effectively suppressing the natural imbalance of the data structure. Conversely, Hierarchical Clustering preserved the structural blockiness and skew introduced by the synthesis method. However, both algorithms suffered from centroid drift and variance distortion in high-dimensional or overlapping environments, confirming that the synthesis process itself struggles to replicate complex multivariate correlations regardless of the downstream algorithm used.

4.2 Comparison with previous Work

This study extends the foundational work of Chen Xinnuo (2025), which established the baseline utility of synthetic data. While previous studies largely focused on supervised learning tasks, such as classification accuracy, or global statistical properties, this research specifically addressed the unsupervised domain. Unlike general utility metrics that average performance over a dataset, our introduction of structural metrics—specifically the Mean Centroid Distance and Gini Coefficient—allowed for a granular diagnosis of why algorithms fail. We demonstrated that utility is not a binary property of the data, but an interaction between the generation method and the analysis method. Specifically, we identified that the orthogonal constraints of CART are fundamentally incompatible with the spherical convexity assumptions of K-Means when the underlying distribution is non-normal.

4.3 Limitations

The scope of this research implies certain limitations that must be considered when interpreting the results. First, we exclusively evaluated the CART method implemented in the synthpop package. Other synthesis techniques, such as parametric methods or deep learning approaches, may produce different geometric artifacts that could interact differently with clustering algorithms. Second, the comparison was strictly limited to K-Means and Agglomerative Hierarchical Clustering. Density-based algorithms like DBSCAN or model-based clustering methods were not tested, though they may offer distinct advantages in handling synthetic noise. Finally, the use of pure Normal and Gamma distributions provided a controlled experimental setting but may not fully capture the chaotic noise, outliers, and mixed-type variables often found in real-world administrative data.

4.4 Generalization

Despite the specific algorithms used, the findings generalize to the broader class of centroid-based versus connectivity-based learning. We posit that any algorithm relying on convex, distance-minimizing objective functions—such as K-Medoids or Linear Discriminant Analysis—will suffer similar degradation when applied to CART-generated synthetic data of skewed distributions. Conversely, methods that prioritize local neighborhood structures, such as Nearest Neighbors or Single-Linkage Clustering, are likely to remain robust to the blocky discontinuities introduced by decision-tree synthesis. This distinction is crucial for data scientists selecting tools for privacy-preserving analytics, as it suggests that the "algorithm fit" is just as critical as the data quality itself.

4.5 Future Research Directions

To further resolve the friction between data privacy and analytical utility, future research should focus on alternative generation methods. Evaluating whether Generative Adversarial Networks (GANs) or Variational Autoencoders (VAEs) can generate smooth, non-orthogonal decision boundaries would be a logical next step to see if they preserve the convexity required for K-Means. Additionally, there is significant potential in developing metric-optimized synthesis loss functions that explicitly penalize

the distortion of centroid distances or cluster variance during the training of the generator. Finally, creating a standardized "Unsupervised Utility Score" based on the structural metrics defined in this study could help automatically flag synthetic datasets that are unsuitable for clustering tasks before they are released to researchers.

4.6 Conclusions

The promise of synthetic data is that it offers "privacy by design" without compromising analytical truth. This research clarifies the boundaries of that promise. We conclude that synthetic data generated via CART is not utility-neutral; it imparts a structural bias that favors connectivity-based algorithms over centroid-based ones.

For organizations leveraging synthetic data, we recommend a clear operational protocol: if the underlying data distribution is unknown or suspected to be skewed, Hierarchical Clustering must be preferred over K-Means to avoid drawing erroneous conclusions from privacy-preserved data. K-Means should be reserved strictly for scenarios where normality can be verified or where the synthesis method is known to preserve spherical convexity. By understanding these geometric interactions, researchers can effectively unlock the value of sensitive data while maintaining rigorous privacy standards.

Appendix A

Appendixes

A.1 Simulation Configuration

The simulation parameters were controlled via a `config.json` file to ensure reproducibility.

```
{
  "simulation": {
    "N": 1000,
    "n": 5,
    "m": 10,
    "random_seed_base": 16
  },
  "parameters": {
    "p": [2, 5, 10],
    "k": [2, 3, 4],
    "separation": [0.1, 2, 6, 10],
    "rho": [0.0, 0.4],
    "distribution": [
      { "name": "normal" },
      { "name": "gamma", "shape": 1.5, "rate": 5.0 }
    ]
  }
}
```